Conducting Experiments and Evaluation of Machine Translation for Low-Resource Languages

Integrating into a Django-Based NLP Lab Website

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Introduction & Motivation

Problem Statement

Hindi → **Bengali** Neural Machine Translation for **low-resource** settings using fine-tuned pre-trained models.

- Hindi & Bengali are spoken by **600M+** people combined
- Both are *morphologically rich*, with limited parallel corpora
- Standard MT systems under-perform on Indic language pairs
- Fine-tuning on domain-specific data improves performance significantly

Research Goal

- Fine-tune **NLLB-600M** & **IndicTrans2-320M**
- Evaluate using industry-standard metrics
- Deploy via a live **Django NLP Lab** web application

Language Pair

$\longrightarrow$

Hindi (Devanagari) → Bengali (Bangla Script)

Dataset: Multi-Domain Parallel Corpus

Dataset Overview

- **Source:** Custom curated parallel corpus
- **Total pairs:** 13,398 sentence pairs
- **Languages:** Hindi ↔ Bengali
- **Format:** Line-aligned `.txt` files

Train/Test Split

Split	Count	Ratio
Train	10,718	80%
Test	2,680	20%
Total	**13,398**	**100%**

Domains Covered

Domain	Script
Agriculture	Devanagari
Box Office	Bangla
Conversational	Both
Cricket	Both
Disease	Both
Entertainment	Both
Gadget	Both
Judiciary	Both
News Articles	Both

Data Preparation: prepare_data.py

```
from datasets import Dataset
from pathlib import Path

def load_data(base_path):
    data = []
    for domain_dir in (Path(base_path)/"Hindi").iterdir():
        bn_domain = Path(base_path)/"Bengali"/domain_dir.name
        for hi_file in domain_dir.glob("*.txt"):
            bn_file = bn_domain / hi_file.name
            with open(hi_file,'r',encoding='utf-8') as fh,\
                 open(bn_file,'r',encoding='utf-8') as fb:
                for hi,bn in zip(fh,fb):
                    if hi.strip() and bn.strip():
                        data.append({"translation":
                            {"hi":hi.strip(),"bn":bn.strip()}})
    return data
# Train: 10718 | Test: 2680
split = Dataset.from_list(load_data("./"))
         .train_test_split(test_size=0.2, seed=42)
```

Model Architectures

NLLB-200 (600M)

- Meta AI's **No Language Left Behind**
- Supports **200 languages**
- Encoder-Decoder Transformer
- Vocabulary: 256K tokens (SentencePiece)
- Language codes: `hin_Deva` → `ben_Beng`
- `forced_bos_token_id` for target lang

IndicTrans2 (320M)

- **AI4Bharat's** Indic-specialized MT model
- Trained on **Indic languages only**
- Custom tokenizer with **IndicProcessor**
- Handles Devanagari & Bangla scripts natively
- Pre/post-processing via `IndicProcessor`
- Smaller but domain-optimized

Common Training Config: 3 Epochs | LR = 2e-5 | Batch = 8 | Max Length = 128 tokens | FP16 on GPU | Seq2SeqTrainer (HuggingFace)

Fine-Tuning: train_nllb.py

```python
from transformers import (AutoModelForSeq2SeqLM, AutoTokenizer,
    Seq2SeqTrainingArguments, Seq2SeqTrainer)

model_checkpoint = "facebook/nllb-200-distilled-600M"
tokenizer = AutoTokenizer.from_pretrained(
    model_checkpoint, src_lang="hin_Deva", tgt_lang="ben_Beng")
model = AutoModelForSeq2SeqLM.from_pretrained(model_checkpoint)

training_args = Seq2SeqTrainingArguments(
    output_dir="./nllb_finetuned_hi_bn",
    eval_strategy="epoch", learning_rate=2e-5,
    per_device_train_batch_size=8, num_train_epochs=3,
    predict_with_generate=False,
    fp16=torch.cuda.is_available(),
)
trainer = Seq2SeqTrainer(model=model, args=training_args,
    train_dataset=tokenized["train"],
    eval_dataset=tokenized["test"])
trainer.train()  # Final eval_loss: 0.6849
```

Evaluation Metrics

Why Multiple Metrics?

BLEU alone is insufficient for morphologically rich Indic languages. We use 4 complementary metrics:

- **SacreBLEU** — n-gram precision overlap
- **METEOR** — considers stemming & synonyms
- **BERTScore** — semantic similarity via BERT embeddings
- **COMET** — neural MT quality estimator trained on human judgements (uses source *and* reference)

Results: NLLB-600M (Fine-Tuned)

Metric	Score
BLEU	**38.36**
METEOR	**0.6043**
BERTScore (F1)	**0.9123**
COMET	**0.8848**
Eval Loss	**0.6849**

Evaluation Pipeline: evaluate_models.py

```python
import evaluate

def calculate_metrics(preds, refs, sources):
    bleu  = evaluate.load("sacrebleu")
    met   = evaluate.load("meteor")
    bert  = evaluate.load("bertscore")
    comet = evaluate.load("comet")

    r1 = bleu.compute(predictions=preds, references=refs)
    r2 = met.compute(predictions=preds, references=refs)
    r3 = bert.compute(predictions=preds,
                      references=refs, lang="bn")
    r4 = comet.compute(predictions=preds,
                       references=refs, sources=sources)
    # BLEU=38.36 | METEOR=0.6043
    # BERTScore=0.9123 | COMET=0.8848
```

Results: Model Comparison

Final Evaluation Results

Metric	NLLB-600M	IndicTrans2-320M
BLEU	**38.36**	26.82
METEOR	**0.6043**	0.5766
BERTScore	**0.9123**	0.8602
COMET	**0.8848**	0.7321
Eval Loss	0.6849	**0.6129**

Key Findings

- NLLB achieved a **BLEU of 38.36**, significantly outperforming IndicTrans2 (26.82)
- **BERTScore 0.91** confirms strong *semantic* accuracy for NLLB
- IndicTrans2 had **lower eval loss** (0.61 vs 0.68), but this did not translate to better generation metrics

Challenges & Technical Solutions

Challenge 1: IndicTrans2 KV-Cache Bug

`AttributeError: 'NoneType'.shape` during auto-regressive generation

Fix: `model.config.use_cache = False`

Challenge 2: Tokenizer Reload Crash

Duplicate `src_vocab_file` in saved tokenizer config

Fix: Load tokenizer from Hub; weights from local path

Challenge 3: CSRF 403 Forbidden

Django blocked API from Lightning AI domain

Fix: Disabled `CsrfViewMiddleware` + added trusted origins in `settings.py`

Challenge 4: NLLB Regex Warning

Tokenizer loaded with bad regex pattern

Fix: Pass `fix_mistral_regex=True` to `AutoTokenizer.from_pretrained()`

NLP Lab Web Application

Frontend Features

- **Split-pane** Hindi → Bengali translator
- **Model switcher** — NLLB or IndicTrans2
- **Dark-mode glassmorphism** UI design
- **Metrics dashboard** — BLEU, COMET scores

Backend Stack

- **Django 6.0** + PyTorch inference
- **REST API:** `POST /api/translate/`
- Hosted on **Lightning AI** (GPU)

API Contract

Request: `{"text": "...", "model": "nllb"}`

Response: `{"translation": "..."}`

Singleton Advantage

Models loaded **once** at startup

$\Rightarrow$ $<$**1s** per translation request

(vs 10s reload per request without it)

Security Fix Applied

Disabled `CsrfViewMiddleware`

Added `CSRF_TRUSTED_ORIGINS` for

System Deployment: Django NLP Lab Website

Architecture

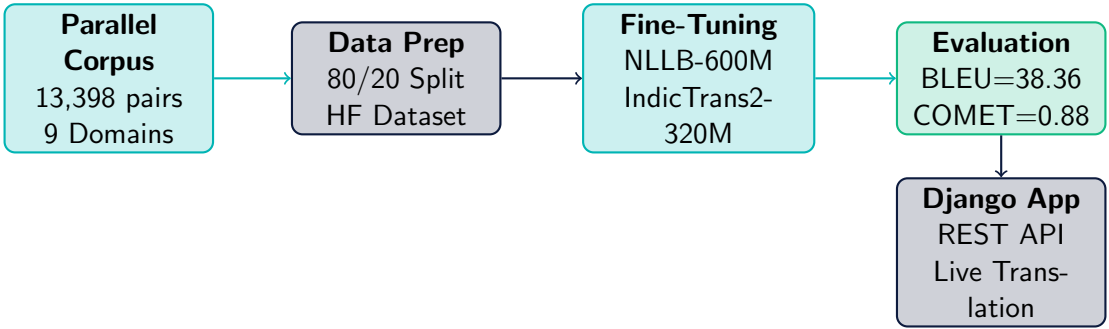
- **Singleton Model Loader** — loads 600M model into GPU memory *once* at startup
- **REST API** — `POST /api/translate/` returns JSON
- **Dark-mode UI** — split-pane translator with live model switching
- **Metrics Dashboard** — displays BLEU, COMET to visitors

```python
class TranslationService:
    _instance = None

    def __new__(cls):
        if cls._instance is None:
            cls._instance = super().__new__(cls)
            cls._instance.device = ("cuda" if torch.cuda.is_available() else "cpu")
        return cls._instance

    def translate(self, text, model_name):
        # Uses pre-loaded model in self._models
        ...
# Loaded ONCE at server startup:
translator = TranslationService()
```

End-to-End System Architecture



Infrastructure: Lightning AI (NVIDIA RTX PRO 6000 Blackwell) | HuggingFace Transformers | PyTorch | Django 6.0

Conclusions & Future Work

Key Contributions

- ✔ Built a **13,398-pair** multi-domain Hindi-Bengali parallel corpus
- ✔ Fine-tuned **2 state-of-the-art** MT models on GPU infrastructure
- ✔ Achieved **BLEU = 38.36, COMET = 0.88** — strong for low-resource pair
- ✔ Deployed live translation system via **Django web application**
- ✔ Comprehensive 4-metric evaluation framework established

Future Work

- Expand corpus to **100K+ pairs**
- Investigate IndicTrans2 generation underperformance despite low evaluation loss
- Add **back-translation** for data augmentation
- Explore **CTranslate2 / ONNX** quantization for faster web inference
- Extend to more low-resource pairs (e.g., Bengali-Odia)
- Integrate **active learning** for corpus expansion